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APZ 3230

PRESSURE TRANSMITTER FOR VERY LOW PRESSURES AND VACUUM

DESCRIPTION

APZ 3230 is based on a silicon piezoresistive sensor without media isolation diaphragm and therefore is capable of measuring very low pressures and vacuum. Compatible with non-aggressive and non-corrosive gases and low viscosity oils, APZ 3230 has a great long-term stability and reduced temperature error comparing to the traditional models with diaphragms. Dry sensor makes this model preferable in applications where media contamination with filling oil is not desirable.

SPECIFICATIONS

Pressure ranges: 6 mbar to 1 bar
Accuracy: up to ±0.25%
Outputs: 4...20 mA (option – Ex ia); 0...20 mA; 0...10 V; 0...5 V; HART®; RS-485 / Modbus RTU
Sensor: silicon piezoresistive, no media isolation
Pressure port: G1/2″; G1/4″; 1/2″ NPT; 1/4″ NPT; M20x1.5; other.
Media temperature: -40…+90 °C
Ambient temperature: -40…+85 °C
Optional: field housing with/without graphics display

APPLICATIONS

Medical equipment
Laboratory instruments
Clean rooms
HVAC
Vacuum applications

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TECHNICAL SPECIFICATIONS

MEASURING RANGES

Pressure range, mbar Gauge	Overpressure, mbar	Burst pressure, mbar	Pressure range, mbar Gauge	Overpressure, mbar	Burst pressure, mbar
0...-1000	3000	5000	0...100	300	500
0...6.0	30	60	0...160	1000	1700
0...10	60	100	0...250	1000	1700
0...16	60	100	0...400	1000	1700
0...25	60	100	0...600	3000	5000
0...40	150	250	0...1000	3000	5000
0...60	150	250			

PERFORMANCE

PERFORMANCE	P > 400 mbar	40 mbar ≤ P ≤ 400 mbar	P < 40 mbar
Accuracy, % of span*	≤ ±0.25	≤ ±0.5	≤ ±1
Temperature effect (% of span / 10 °C)	≤ ±0.1	≤ ±0.15	≤ ±0.2
Compensated range	0…+50 °C	0…+50 °C	0…+50 °C
Power supply effect	≤ ±0.05% of span / 10 V		
Load resistance effect	≤ ±0.05% of span / kOhm		
Long-term stability	≤ ±0.2% of span / year		
Response time (10...90%)	≤ 1 ms with analog output, ≤ 200 ms with digital output		

* Accuracy includes non-linearity, hysteresis and non-repeatability.

OPERATING CONDITIONS

Medium temperature (depends on seal)	-40...+90 °C		
Ambient temperature	-40...+85 °C		
Storage temperature	-40...+85 °C		
Approval	0Ex ia IIC T6…T4 Ga X		
Temperature class	T4	T5	T6
Ambient temperature	-40…+80 °C	-40…+60 °C	-40…+50 °C
Vibration resistance	10 g RMS, 20–2000 Hz		
Shock resistance	100 g / 11 ms		
Service life	> 100 x 10^6 cycles		

MECHANICAL SPECIFICATIONS

Pressure port material	stainless steel 304 (1.4301) standard; 316L (1.4404) - optional			
Housing material	stainless steel 316L (1.4404)			
Seal	EPDM (-40…+90 °C); NBR (-25…+90 °C); FKM (-25…+90 °C);			
Diaphragm	silicon, pyrex, RTV			
Wetted parts	Diaphragm, pressure port, seal			
Pressure port	G 1/2″ DIN 3852 / EN 837	G 1/4″ DIN 3852 / EN 837	1/2″ NPT	1/4″ NPT
	M20x1.5 DIN 3852 / EN 837	M16x1.5 DIN 3852 / EN 837	M12x1.5 DIN 3852 / EN 837	
	M12x1.25 DIN 3852 / EN 837	M12x1 DIN 3852 / EN 837	M10x1 DIN 3852	

Electrical connection	Ingress protection	Cross section	Cable diameter
DIN 43650A (4 pin)	IP65	1.5 mm²	6…8 mm
Binder 723 (5 pin)	IP67	0.75 mm²	6…8 mm
M12x1 (5 pin)	IP67	0.75 mm²	6…8 mm
Buccaneer (4 pin)	IP68	1.5 mm²	6…8 mm
Cable gland, M12x1.5	IP67	0.14 mm²	5 mm
Cable gland, stainless steel	IP68	0.14 mm²	7.5 mm
Field housing, cable gland M20x1.5	IP67	1.5 mm²	7...10 mm

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DIGITAL DISPLAY (only for field housing version)

Display type	OLED 128x64 pixels (30x16 mm)
Displayed units	bar, mbar, MPa, kPa, Pa, psi, mmHg, mWc, ftH2O, %, mA, user
Displayed values range	-1999…9999
Display accuracy	0.1 % of span ± 1 digit
Settling time	< 1 s (with damping disabled)
Damping	0.3…30 s (programmable)

ELECTRICAL SPECIFICATIONS

Output signal	Power supply, U_s	Load resistance, R	Power consumption
4…20 mA / 2-wire	12…36 V 18…42 V (with display)	≤ [(U_s – 12 V) / 0.02 A] Ohm* ≤ [(U_s – 18 V) / 0.02 A] Ohm* (with display)	≤ 26 mA
4…20 mA / HART®			
4…20 mA / 3-wire	12…36 V	≤ 500 Ohm	
0…20 mA / 3-wire			
0…10 V / 3-wire		≥ 10 kOhm	< 7 mA
0…5 V / 3-wire		≥ 5 kOhm	
0.5…4.5 V / 3-wire	5 V		≤ 2 mA
0.5…4.5 V / 3-wire	6…15 V		≤ 7 mA
RS-485 / Modbus RTU	12…36 V	-	≤ 7 mA

* For 4…20 mA / HART® output signal, minimum load resistance for digital communication: 250 Ohm.

Safe values for intrinsically safe design 0Ex ia IIC T6…T4 Ga X:

Parameter	2-wire	3-wire, 4-wire
Maximum voltage, Ui	28 V	6 V
Maximum current, Ii	93 mA	60 mA
Maximum power, Pi	660 mW	100 mW
Maximum internal inductance, Li	10 µH	10 µH
Maximum internal capacitance, Ci	15 nF	500 nF

ELECTRICAL CONNECTIONS / PIN ASSIGNMENT

Circuits		DIN 43650	M12x1	Binder 723	Buccaneer	Cable gland	Field housing with M20x1.5 cable gland
2-wire	power +	1	1	3	1	white	2
	power -	2	2	4	2	brown	3
	shield	GND	4	5	4	yellow-green	1
3-wire	power +	1	1	3	1	white	2
	power -	2	2	4	2	brown	3
	signal +	3	3	1	3	green	4
	shield	GND	4	5	4	yellow-green	1
RS-485 4-wire	power +	-	3	3	-	white	-
	power -	-	1	1	-	brown	-
	A	-	4	4	-	yellow	-
	B	-	5	5	-	green	-
	shield	-	2	2	-	yellow-green	-

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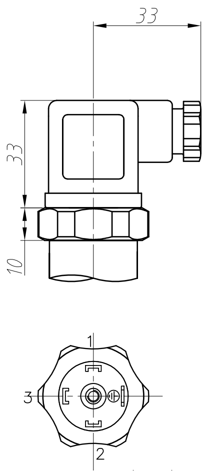
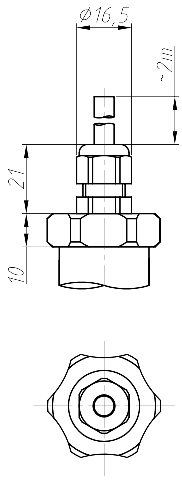
ELECTRICAL CONNECTIONS, DIMENSIONS (mm)

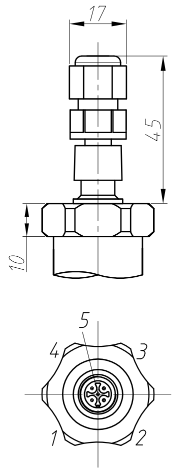
DIN 43650A (IP65)



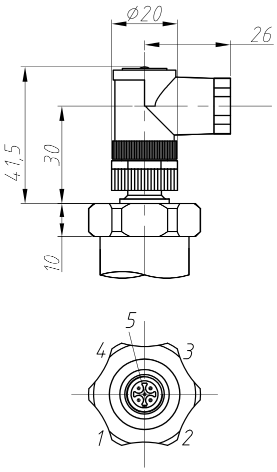
Cable gland M12x1.5 (IP67)



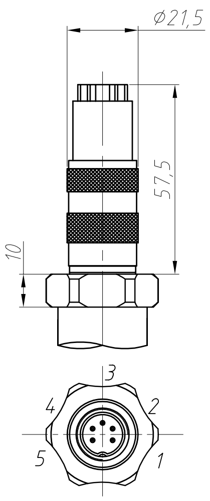
M12x1 straight connector (IP67)



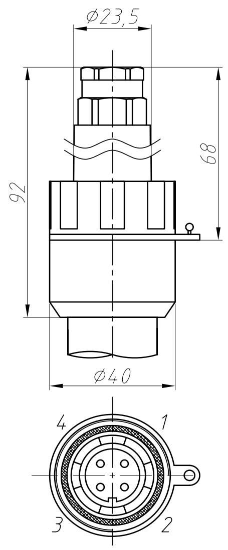
M12x1 angular connector (IP67)



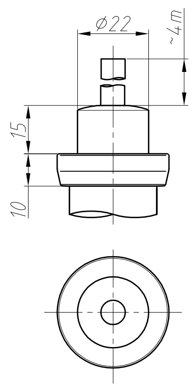
Binder 723 (IP67)



Buccaneer (IP68)

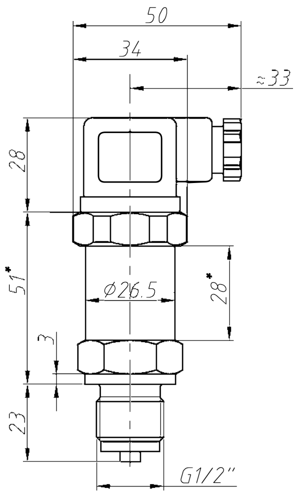


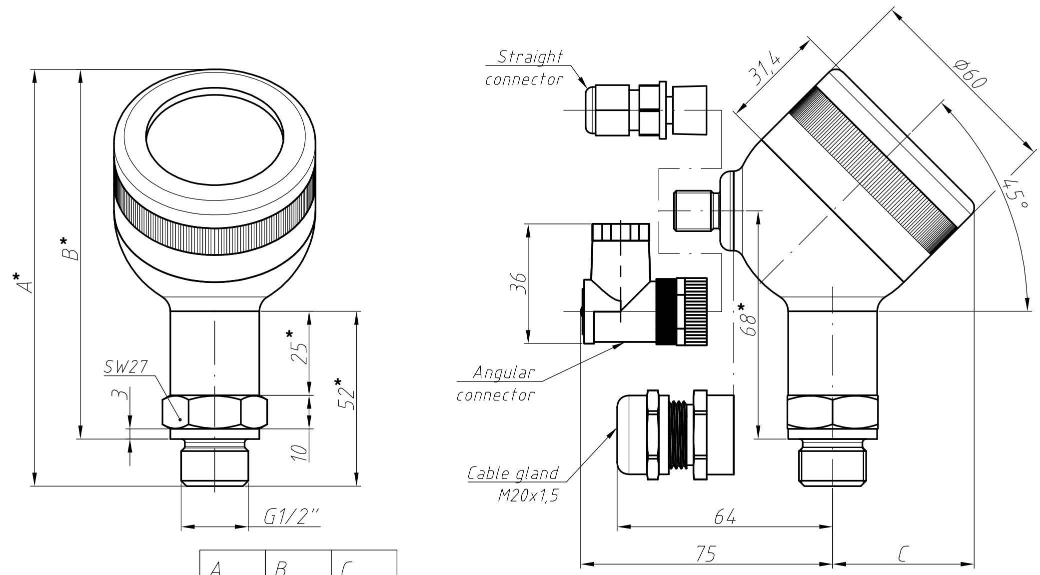
Stainless steel cable gland (IP68)



DIMENSIONS (mm)

Standard	Field housing





	A	B	C
with display	124	110	42
without display	121	107	39

* Housing of Ex ia version is 25 mm longer.
Housing of pressure transmitter with RS485 / ModbusRTU output signal is 34 mm longer.
Housing of pressure transmitter with HART® output signal is 42 mm longer.

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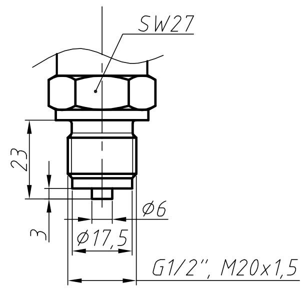
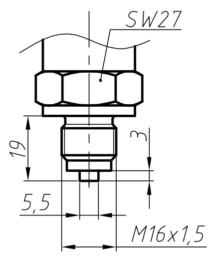
PRESSURE PORTS, DIMENSIONS (mm)

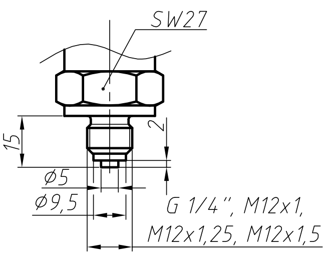
M20x1.5; G1/2″ EN 837



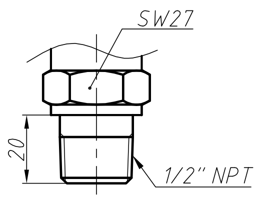
M16x1.5 EN 837



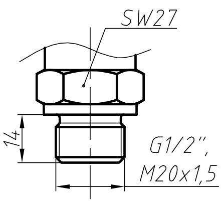
G1/4″; M12x1; M12x1.25 M12x1.5 EN 837



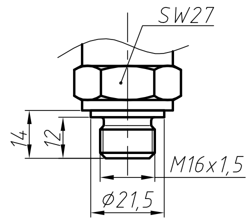
1/2″ NPT



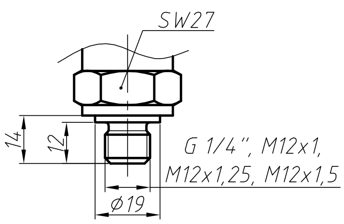
M20x1.5; G1/2″ DIN 3852



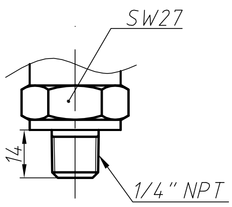
M16x1.5 DIN 3852



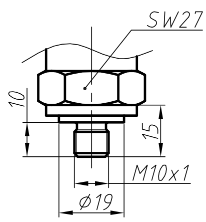
G1/4″; M12x1; M12x1.25 M12x1.5 DIN 3852



1/4″ NPT



M10x1 DIN 3852



ORDERING CODE

APZ 3230	-X	-X	-XXXX	-X	-XX	-X	-XXX	-X	-XX

MEASUREMENT TYPE

	-X
Gauge	G
Vacuum, LRL = -1000 mbar	V

UNIT OF MEASUREMENT

	-X
mbar	R
kPa	K
Other (specify when ordering)	X

UPPER RANGE LIMIT (URL)

mbar		kPa	
6.0	6000	0.6	0600
10	1001	1.0	1000
16	1601	1.6	1600
25	2501	2.5	2500
40	4001	4.0	4000
60	6001	6.0	6000
100	1002	10	1001
160	1602	16	1601
250	2502	25	2501
400	4002	40	4001
600	6002	60	6001
1000	1003	100	1002
Other	XXXX	Other	XXXX

ACCURACY

	-X
0.25% (P > 400 mbar) (standard)	C
0.5% (40 mbar ≤ P ≤ 400 mbar) (standard)	D
1% (P < 40 mbar) (standard)	E
Other (specify when ordering)	X

ELECTRICAL CONNECTION

	-XX
DIN 43650A	10
Binder 723	20
M12x1, straight connector	30
M12x1, angular connector	31
Cable gland M12x1.5 + cable 2 m	40
Stainless steel cable gland + cable 4 m	41
Buccaneer	50
Field housing without display, cable gland M20x1.5	60
Field housing with display, cable gland M20x1.5	67
Field housing with display, straight connector M12x1	64
Field housing with display, angular connector M12x1	65
Other (specify when ordering)	XX

OUTPUT SIGNAL

	-X
4…20 mA / 2-wire (standard)	A
4…20 mA / 2-wire, 0Ex ia IIC T6…T4 Ga X	Q
4…20 mA / 3-wire	B
0…20 mA / 3-wire	C
0…10 V / 3-wire	D
0…5 V / 3-wire	E
0.5…4.5 V / 3-wire, U_S = 5 V, 0Ex ia IIC T6…T4 Ga X	R
0.5…4.5 V / 3-wire, U_S = 6...15 V	K
RS-485 / Modbus RTU	M
4…20 mA / HART®	H
Other (specify when ordering)	X

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ORDERING CODE (CONTINUED)

APZ 3230	-X	-X	-XXXX	-X	-XX	-X	-XXX	-X	-XX
PRESSURE PORT									
M20x1.5 DIN 3852 (standard)							200		
M20x1.5 EN 837 (standard)							201		
G1/2″ DIN 3852 (standard)							720		
G1/2″ EN 837 (standard)							721		
G1/4″ DIN 3852 (standard)							740		
G1/4″ EN 837							741		
M16x1.5 DIN 3852							160		
M16x1.5 EN 837							161		
M12x1.5 DIN 3852							122		
M12x1.5 EN 837							123		
M10x1 DIN 3852							100		
M12x1 DIN 3852							120		
M12x1 EN 837							121		
1/4″ NPT							840		
1/2″ NPT							820		
Other (specify when ordering)							XXX		
SEALS									
FKM (-25…+90 °C, standard)								F	
NBR (-25…+90 °C)								N	
EPDM (-40…+90 °C)								E	
Other (specify when ordering)								X	
VERSION									
Standard									00
Zero trim (requires ZCON 100 configurator)									01
Compound filled version									16
Other (specify when ordering)									XX

Example: APZ 3230-G-R-4001-D-10-A-100-F-00

ACCESSORIES

				
DZ 10 Pressure snubber	ZCON 100 Zero trim and range selection device	ANZ 200 Plug-in display for transmitters with 4-20 mA output	PZ 1024 Power supply unit	BZ 05 / BZ 10 Dry air junction box for submersible transmitters